

END TO END LTSpice SIMULATION FLOW

Stage	Files/Components Involved	Purpose / Role
1. Schematic Drawing	.asc (schematic file)	You draw your circuit in LTspice GUI using symbols and components. Saved as .asc file.
2. Component Definition	.asy (symbol files)	Each component like resistor, capacitor, opamp, etc. uses a .asy symbol file that defines its graphical view.
3. Netlist Generation	Netlist (internal), optionally viewable via View > SPICE Netlist	LTspice converts .asc into a SPICE netlist: a plain-text description of all connections and models.
4. Model & Library Linking	.lib, .sub, .mod (Model files)	Custom components (BJTs, MOSFETs, etc.) require model files (device equations, parasitics, etc.).
5. Parsing & Compilation	Internal SPICE Engine	LTspice parses the netlist, checks all connections and component parameters, compiles the simulation structure.
6. Simulation Execution	.net file (intermediate) .log file (optional output)	.net contains a simulation topology and setup. .log gives textual info like warnings, errors, OP data.
7. Data Output Generation	.raw (binary waveform file)	After running, voltage, current, and other waveforms are written in .raw — the data you see in the graph.

- **.asc** – Main design schematic with node connections and component values.
- **.asy** – Symbol files used for drawing. These are not electrically functional but graphical.
- **.net** – Internal topology used during simulation (not editable).
- **.lib / .sub / .mod** – SPICE model definitions (behavioral/equation-based).
- **.raw** – Binary file storing simulation waveforms like V(n001), I(R1), etc.
- **.log** – Text output showing simulation summary, errors, and operating point values.

1. **Run**
2. **.asc** is parsed → creates a SPICE netlist
3. Symbol **.asy** files are referenced for component mapping
4. **SPICE models** are loaded (**.lib**, **.mod**)
5. Simulation is compiled (**.net**) and executed
6. Output waveforms are stored in **.raw**
7. Results are visualized on the waveform viewer